

Milestone 02 – Improve Script and Metadata Documentation

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1 Overview of v1.5.1

This milestone focuses on strengthening project transparency and reproducibility by embedding documentation within analysis folders and formalizing metadata management.

1.1 Goal

Enhance clarity, reproducibility, and maintainability of all project scripts by standardizing documentation headers, comments, and auxiliary files.

2 Project Structure

```
FFA_regional-skew/
.gitignore           # Prevents sensitive/local files from being pushed
arcgis_project/      # ArcGIS Pro .aprx project and supporting layers
CHANGELOG.Rmd        # Human-readable changelog (semantic versioning)
data/                # Raw, processed, meta, intermediate, and QA data
docs/                # Documentation, workflow notes, metadata, checklists
FFA_regional-skew.Rproj # RStudio project file (keep in root)
notebooks/           # Ad hoc .Rmd/.qmd experiments (empty or minimal)
notes/               # Team notes, meeting logs, brainstorm
R/                   # Analysis scripts (milestone-organized)
  01_download/       # Download process raw data
  02_clean_validate/ # QA scripts and validation checks
  03_covariates/      # Covariate extraction
  04_modeling/        # Statistical modeling (GAMs, Elastic Net, etc.)
  06_eval/            # Model diagnostics, residuals, validation
  utils/              # Reusable functions (organized by domain)
README.md            # GitHub-readable overview and navigation aid
README.Rmd           # Editable RMarkdown version
reports/             # Knitted .Rmd/.qmd milestone reports
results/             # Manuscript-ready figures, models, and outputs
sandbox/             # Staging area for review
```

3 v1.5.1 – Improve Script and Metadata Documentation

4 v1.5.1 Tasklist

Step	Task	Status
2.1	Standardize header blocks across R scripts	[X]
2.2	Add README.md to each R/ subfolder	[X]
2.3	Ensure all scripts are self-contained	[X]
2.4	Review document dependencies	[X]
2.5	Review docs/metadata covariates	[X]
2.6	Standardize use of .Rmd and .md for all folders	[X]
2.7	Develop log_README.md templates to include in R/ and data/	[X]
2.8	Link README-style metadata to milestone logs	[X]
2.9	Commit and push changes as tag v1.5.3-docs-enhancement	[X]

4.1 Description of Tasks Completed or In Progress

4.1.1 2.1 Standardize header blocks across all R scripts

This milestone relates to the following folders in R/01_downloads, R/02_clean_validate, and R/utils. Check the following are included in all scripts: - Script Name line 20 - Author: line 20 - Date Created line 20 - Last Updated line 20 - Change Log new line bulleted - Purpose line 20 - Workflow Summary new line numbered - Input/Data URLs new line bulleted - Outputs new line bulleted – add subfolder - Dependencies new line bulleted – description on line 20 - Helper Functions new line bulleted - Related Milestone Reports new line bulleted - Move notes to Move notes to notes/script-notes_ and _developer-log - Make a consistent use of ===== and -----

4.1.2 2.2 Add README files to each major script directory

The README should include the following:

- High level workflow summaries
- A highlight of important scripts and expected outputs
- Links to data directories and relevant logs

4.1.3 2.3 Ensure all scripts are self-contained:

- Use `here()` for all file paths
- Clearly load libraries at top
- Minimize hard-coded paths or variables

4.1.4 2.4 Review Document dependencies:

- Ensure that each script lists required packages
- Consider using a `renv` or `DESCRIPTION` file for reproducibility
- Add clarifying comments to non-obvious code
- Remove commented-out legacy code unless justified
- Cross-reference milestones or related scripts at the top of each file

4.1.5 2.5 Review docs/metadata covariates

The file `docs/data_dictionary.pdf` summarizes `docs/metadata/descriptions/covariate_metadata_vxx.csv`

Check the following items exist: - Variable names - Descriptions - Units - Data types - Possible value ranges
- Notes about relevance, quality, or source

4.1.6 2.6 Standardize use of .Rmd and .md for all folders

1. Define Purpose for Each Format

- `.Rmd` = Source file for dynamic documents with embedded R code (used for knitting reports, README files, summaries).
- `.md` = Output file (or static version) typically created by knitting an `.Rmd` or manually written for simple documentation (e.g., GitHub READMEs).

2. Audit Each Folder Check folders like `docs/`, `data/`, `notes/`, `reports/`:

- Is there an `.Rmd` file with analysis, notes, or a README?
- Is there a matching `.md` file? If yes, is it auto-generated?
- Is the `.md` up to date with the `.Rmd`?

3. Standardize Conventions

- Every folder should have a `README.Rmd` with dynamic content (e.g., script summary, table of outputs).
- Then, knit it to `README.md` for GitHub display or static viewing.

4. Clean Up Inconsistencies

- Remove stale `.md` files that aren't backed by an `.Rmd`.
- Rename files for consistency (e.g., avoid both `readme.md` and `README.md`).
- Ensure every `.Rmd` can knit cleanly and reproducibly to `.md`.

4.1.7 2.7 Develop log_README.md templates to include in R/ and data/

Create standardized log templates (log_README.md) for documenting changes, decisions, and file-level notes within the project R/ and data/ directories.

The purpose is to act like a local changelog or activity log specific to that folder to capture:

- Script or file additions/removals
- Data transformations or validation steps
- Notes about known issues, exceptions, or edits
- Author/date stamps for transparency
- Help maintain folder-level traceability for collaborative or long-term projects

5 Next Steps

6 Summary

This milestone strengthens reproducibility by formalizing internal documentation practices. It includes README templates for R/ subfolders, a planned data dictionary, and introduces log_README.md documentation inside each milestone to clarify roles, assumptions, and outputs.

6.0.1 Reproducibility Notes

- See DESCRIPTION and renv.lock for project-wide dependencies and package versions.